Annexure A – Article Review Report

ARTICLE REVIEW REPORT

Article: “Reinforcement learning assisted oxygen therapy for COVID-19 patients under intensive care” Authors: Hua Zheng, Jiahao Zhu, Wei Xie, Judy Zhong Journal: BMC Medical Informatics and Decision Making Year: 2021 DOI: 10.1186/s12911-021-01712-6 Publisher: Springer Nature / BMC

Abstract This report reviews a 2021 peer-reviewed research article applying deep reinforcement learning to personalized oxygen therapy for critically ill COVID-19 patients. The paper models patient health trajectories as a Markov decision process and uses Deep Deterministic Policy Gradient to recommend a continuous oxygen-flow rate.

Main contribution The method learns from longitudinal electronic health records and treats survival as the long-term objective. The study reports a mean mortality of 5.37% under the RL policy versus 7.94% under standard care, alongside a lower average recommended oxygen flow.

Critical assessment The results are promising but retrospective. Treatment decisions in historical EHR data are confounded by clinician judgment and patient severity. Cross-validation does not replace external or prospective validation. Off-policy evaluation, uncertainty estimation, subgroup fairness analysis and multi-hospital testing would strengthen the evidence.

CO4 relevance The work demonstrates MDPs, continuous actions, reward design, Q-function approximation and policy learning. It is a direct application of Reinforcement Learning to healthcare decision support.

Ethical conclusion The model should be treated as decision support, not autonomous treatment. Clinical safety constraints, explainability, privacy, bias evaluation, regulatory review and clinician oversight are necessary.